

Madhur Gupta

Full Stack Developer

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PROFESSIONAL EXPERIENCE

Qen Labs

Feb 2024 – Present

Full Stack Developer

Remote

- Built a high-throughput geospatial visualization engine using **Kepler.gl** and **deck.gl** to render datasets exceeding **1M+ spatiotemporal points**, leveraging viewport-driven rendering to reduce client memory overhead by **40%**.
- Engineered dynamic map tools and automated data ingestion pipelines for **GeoJSON** and **H3** vector tiles across 10+ complex layers.
- Integrated AI workflows (**Claude**) into routine development for component scaffolding, test coverage automation, and system refactoring, boosting team sprint output by **30%**.
- Centralized global state management using **Redux** and decoupled monolithic frontend views into reusable **TypeScript** hooks, significantly lowering technical debt.
- Designed secure cloud data pipelines to fetch assets from **Google Cloud Storage** and deployed containerized microservices on **GCP Cloud Run** with **OAuth 2.0** security.
- Streamlined environment setups by containerizing backend services with **Docker**, cutting build times by **50%** across the GCP ecosystem.

Professos

Jun 2023 – Aug 2023

Full Stack Intern

Remote

- Developed core features for a **MERN stack** recruitment portal designed to match job seekers with roles based on candidate skill profiles.
- Re-architected frontend user layouts with **Tailwind CSS**, improving page load speed, component modularity, and overall UI responsiveness.

KEY PROJECTS

Algorithm Visualizer & Multiplayer Platform

visualiz.vercel.app

React, Node.js, Socket.io, Tailwind CSS

Ongoing

- Built a real-time WebSocket communication layer with **Socket.io** supporting sub-100ms synchronization latency for **50+ concurrent users** in turn-based sessions.
- Designed interactive step-by-step visualizers for Graph (Dijkstra, A*) and Sorting algorithms, maintaining a smooth **60fps execution** through optimized custom **React Hooks**.
- Implemented a deterministic seed-based generator for Sudoku boards, producing unique, shareable game IDs and 100% reproducible board states.

Budget Management App

MERN Stack, MongoDB, Node.js

- Developed a full-stack expense tracking platform featuring automated recurring bill reminders and dynamic salary dashboards using indexed MongoDB queries.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, SQL (PostgreSQL), C++, C

Frontend: React, Redux, Next.js, Deck.gl, Kepler.gl, Tailwind CSS

Backend & Cloud: Node.js, Express, PostGIS, GCP (Cloud Run), Docker, MongoDB, Socket.io

Developer Tools: Git, Linux, Postman, VS Code, Claude Code, Cursor

EDUCATION

Indian Institute of Information Technology (IIIT), Kota

Dec 2020 – June 2024

Bachelor of Technology in Electronics and Communication

Kota, Rajasthan

ACHIEVEMENTS & RESEARCH

- ACM Publication:** Lead Author of "MapYog - Intelligent Spatiotemporal Data Explorer" (2024), presenting research on high-scale spatial data visualization.
- Problem Solving:** Solved **1000+ algorithmic problems** across LeetCode and Codeforces focusing on Data Structures, Dynamic Programming, and Graph Algorithms.
- GATE CS 2024:** Qualified the GATE Computer Science exam, demonstrating strong foundations in core CS subjects.
- Hackathons:** Developed a geolocation-verified facial attendance prototype during Codeshastra 8.0 with real-time API verification.